Case Studies in Operations Management

Mathematical Model

Business and Logistics Plan for "Fresh Veggie" (FV)

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1 Introduction

2 Problem Description

3 Model Assumptions

4 Sets

- CD: set of candidate distribution centers.
- DS: set of candidate dark stores.
- $MDCP$: set of candidate parking locations for mobile distribution centers.
- P: set of ports:
$$P = \{\text{San Antonio}, \text{Valparaíso}\}$$
- A: set of airports:
$$A = \{\text{Arturo Merino Benítez International Airport}\}$$
- WM: set of wholesale markets:
$$WM = \{\text{La Vega Central}, \text{Lo Valledor}\}$$
- S: set of supplier countries:
$$S = \{\text{US}, \text{Mexico}, \text{Peru}, \text{Ecuador}, \text{Bolivia}, \text{Argentina}, \text{Spain}, \text{Chile}\}$$
- K: set of products:
$$K = \{\text{Roma Tomatoes}, \text{Cauliflower}, \text{Broccoli}, \text{Asparagus}, \text{Green Bell Pepper}\}$$
- M: set of months in the planning horizon:
$$M = \{\text{Jan}, \text{Feb}, \text{Mar}, \text{Apr}, \text{May}, \text{Jun}, \text{Jul}, \text{Aug}, \text{Sep}, \text{Oct}, \text{Nov}, \text{Dec}\}$$
- R: set of transportation modes:
$$R = \{\text{sea}, \text{air}\}$$
- I: set of facilities considered in the coverage model, where
$$I \subseteq CD \cup DS \cup MDCP$$

5 Parameters

5.1 Distance Parameters

All distances are measured in kilometers (km).

- $d_{ij}^{CD,DS}$: distance from distribution center $i \in CD$ to dark store $j \in DS$.
- d_{ij}^{MDCP}: distance between mobile distribution center parking locations $i, j \in MDCP$.
- $d_{ij}^{CD,P}$: distance from distribution center $i \in CD$ to port $j \in P$.
- $d_{ij}^{CD,A}$: distance from distribution center $i \in CD$ to airport $j \in A$.
- $d_{ij}^{CD,WM}$: distance from distribution center $i \in CD$ to wholesale market $j \in WM$.

5.2 Coverage Parameters

- $z = (z_i)_{i \in I}$: binary tuple representing a coverage combination, where $z_i \in \{0, 1\}$ indicates whether facility $i \in I$ is active in the combination.
- $p(z)$: number of households (population) covered exactly by combination z.

5.3 Cost Parameters

- $c_{k,s,r}^{buy}$: purchase cost (USD/kg) of product $k \in K$ from supplier $s \in S$ using transport mode $r \in R$.
- $c_{k,w}^{WM}$: purchase cost (USD/kg) of product $k \in K$ at wholesale market $w \in WM$.
- v_k: selling price (USD/kg) of product $k \in K$.

5.4 Availability Parameters

- $a_{k,s,t}$: binary parameter equal to 1 if product $k \in K$ is available from supplier $s \in S$ in period $t \in T$.
- $g_{k,r}$: binary parameter equal to 1 if product $k \in K$ can be transported using mode $r \in R$.
- $u_{s,p}^{P}$: binary parameter equal to 1 if supplier $s \in S$ can ship through port $p \in P$.
- $u_{s,a}^{A}$: binary parameter equal to 1 if supplier $s \in S$ can ship through airport $a \in A$.

5.5 Transportation Parameters

- $t_{s,p,r}^{first}$: transportation time (hours) from supplier $s \in S$ to entry point $p \in P \cup A$ using mode $r \in R$. For domestic suppliers ($s \in$ Chile), this value is equal to 0.
- α_r^{LH}: long-haul travel time coefficient (hours/km) for transportation mode $r \in R$.

- $t^{LH}_{i,j,r}$: long-haul transportation time (hours) between locations i and j, computed as:

$$t^{LH}_{i,j,r} = \alpha^{LH}_r \cdot d_{i,j}$$

- α^{city}: urban travel time coefficient (hours/km) for local distribution.
- $t^{city}_{i,j}$: urban transportation time (hours) between facilities or to customers, computed as:

$$t^{city}_{i,j} = \alpha^{city} \cdot d_{i,j}$$

5.6 Demand Parameters

- q_k: demand rate (kg/person/week) of product $k \in K$.

6 Decision Variables

7 Objective Function

8 Constraints

9 Extensions and Special Considerations

10 Data and Calibration

11 Computational Implementation

12 Results

13 Analysis and Discussion

14 Conclusions

15 References

16 Appendices